Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

***Ringkasan*—This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60**

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

amssymb,amsthm,amsmath,array graphicx [caption=false,font=footnotesize]subfig xspace [sortcompress, numbers]natbib stmaryrd xcolor mathtools float textcomp

***Ringkasan*—Abstract should contain approximately 100 to 150 words.**

I. Introduction

Please follow the outlined formatting when submitting your manuscript to the International Symposium on Computational Sensing (ISCS). This section provides guidelines for abstract submissions for all types of presentations in ISCS, *i.e.,* platform, poster, and show-and-tell presentations. Specific requirements for show-and-tell abstracts are highlighted in Sec. II.

A. Paper length

The length of the paper must be up to 2 pages for technical content, figures, and references, and one optional third page containing only references.

B. Font size

Use a font size that is no smaller than 9 points throughout the paper, including figure captions.

C. Bibliography

List and number all bibliographical references at the end of the paper. The references can be numbered in alphabetic order or in order of appearance in the document. When referring to them in the text, type the corresponding reference number in square brackets as shown at the end of this sentence [**?**].

II. Specific Guidelines for Show-and-Tell Abstract

The extended abstract for show-and-tell demos must follow the aforementioned format. The manuscript should

- include pictures of the set-up;
- include a clear explanation of the targeted application and computational method developed;
- include a representative example of acquisitions using the set-up and its associated processing.

III. Conclusion